

Gunhee Shin

PH.D. STUDENT, URBAN ROBOTICS LAB, KAIST

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Education

Ph.D. Student, Electrical Engineering — KAIST

2025 – Present

- Advisor: Prof. Hyun Myung — Urban Robotics Lab

M.S., Electrical Engineering — KAIST

2023 – 2025

- GPA: 3.7/4.3
- Advisor: Prof. Hyun Myung — Urban Robotics Lab
- Thesis: "Uncertainty-aware Multiple LiDAR-Inertial Odometry for Outdoor Mobile Robots"

B.S., Electrical Engineering — INHA UNIVERSITY

2017 – 2023

- GPA: 3.9/4.5
- Thesis: "A Fast Correspondence Matching for Iterative Closest Point Algorithm"

Research Interests

I work on localization for field robots. My focus is LiDAR-inertial odometry that stays accurate under fast motion and vibration, and global localization that bounds drift over large outdoor missions without loop closure or GNSS. I build these methods into complete systems and test them on legged and wheeled robots in challenges and industry projects.

Publications

JOURNAL ARTICLES

1. **G. Shin**, J. Choi, and K. Kim. "A Fast Correspondence Matching for Iterative Closest Point Algorithm." *J. Korea Robot. Soc.*, vol. 17, no. 3, pp. 373–380, 2022.

CONFERENCE PAPERS

1. **G. Shin**, Y. Kim, C. Lee, W. Kim, M. Lee, S. Han, J. Woo, H. Chin, M. Park, and H. Myung†. "Online Target-less Radar-LiDAR-Camera Extrinsic Calibration via Joint Optimization." in *Proc. Int. Conf. Control, Automat. Syst.*, 2026.
2. **G. Shin**, S. Lee, J. Kong, Y.-W. Seo, and H. Myung†. "SE(3)-LIO: Smooth IMU Propagation With Jointly Distributed Poses on SE(3) Manifold for Accurate and Robust LiDAR-Inertial Odometry." in *Proc. IEEE Int. Conf. Robot. Automat.*, 2026. [arXiv:2603.16118](#) · [project page](#)
3. H. Lim, D. Kim, **G. Shin**, J. Shi, I. Vizzo, H. Myung†, J. Park†, and L. Carlone. "KISS-Matcher: Fast and Robust Point Cloud Registration Revisited." in *Proc. IEEE Int. Conf. Robot. Automat.*, 2025. [arXiv:2409.15615](#)
4. **G. Shin**, S. Lee, M. Oh, D. Lee, J. Lee, Y. Seo, and H. Myung†. "NM-LIO: Multiple LiDAR-Inertial Odometry Addressing LiDAR Measurement Noise Discrepancy." in *Proc. Int. Conf. Robot. Intell. Tech. Appl.*, 2024.
5. M. Oh, **G. Shin**, S. Jang, S. Lee, D. Lee, W. Song, B. Yu, H. Lim, J. Lee, and H. Myung†. "B-TMS: Bayesian Traversable Terrain Modeling and Segmentation Across 3D LiDAR Scans and Maps for Enhanced Off-Road Navigation." in *Proc. IEEE Intell. Veh. Symp. Workshops*, 2024. [arXiv:2406.18138](#)
6. H. Lim, K. Han, **G. Shin**, G. Kim, S. Hong, and H. Myung†. "ORORA: Outlier-Robust Radar Odometry." in *Proc. IEEE Int. Conf. Robot. Automat.*, 2023.

† Corresponding author

Patents

1. H. Myung, **G. Shin**, S. Lee, and J. Kong. "LiDAR and Inertial-Based Position and Pose Estimation System and Method." Korean Patent Application No. 10-2026-0080491, May 2026.

Open-Source Software

SE(3)-LIO — github.com/url-kaist/se3-lio

2026

SE(3)-LIVOM — github.com/url-kaist/se3-livom-comfort · [video](#)

2026

Honors & Awards

- Winner, COMFORT Localization Challenge** LEAD — IROS 2026, PITTSBURGH, USA

2026

 - Challenge: Estimate the trajectory of an ANYmal legged robot in six field scenes of the GrandTour dataset.
 - Role: Built the whole system: LiDAR-visual-inertial odometry and online drift correction.
- Winner, Nothing Stands Still Challenge** — CVPR 2026, DENVER, USA

2025

 - Challenge: Align scans of a building site taken at different construction stages into one map.
 - Role: Graph optimization that aligns the scans across stages.
- 2nd Place, Quadruped Robot Challenge** — ICRA 2025, ATLANTA, USA

2025

 - Challenge: Legged robots cross rough-terrain lanes and complete inspection and manipulation tasks.
 - Role: LiDAR-inertial odometry for onboard navigation.
- 2nd Place, Quadruped Robot Challenge (Autonomous Track)** — ICRA 2024, YOKOHAMA, JAPAN

2024

 - Challenge: Legged robots run as many laps of a course as they can in 10 minutes, fully autonomously.
 - Role: Localization in a prebuilt map.
- Winner, HILTI SLAM Challenge (LiDAR Single-Session)** — ICRA 2023, LONDON, UK

2023

 - Challenge: Estimate accurate trajectories from data recorded on construction sites, in narrow rooms, and in the dark.
 - Role: LiDAR-inertial odometry and merging of multiple sessions.

Projects

- GNSS-Free Visual-Inertial Navigation for Drones** — HANWHA AEROSPACE

2026 – Present

 - Role: Fused visual-inertial odometry with satellite-image localization.
- Online Target-less Radar-LiDAR-Camera Calibration** LEAD — LG INNOTEK

2025 – Present

 - Role: Led development of online calibration between radar, LiDAR, and camera without targets.
- Off-Road Autonomous Driving in Unstructured Environments (Autonomy Hub)** — HANWHA AEROSPACE

2023 – 2026

 - Role: Built LiDAR-inertial odometry (SE(3)-LIO) and fused it with satellite-image localization for driving without GNSS.
- Mobility Intelligence Software for Quadruped Robots** — KEIT

2022 – 2026

 - Role: Set up the multi-LiDAR sensor system and built odometry and localization in prebuilt maps.
- Semantic SLAM for Spatial AI** — LG ELECTRONICS

2023

 - Role: Ran 3D scene graph SLAM (Hydra) on real sensor data and added object-level loop closing.

Skills

Languages	C++, Python
SLAM / Optimization	ROS, GTSAM, Ceres, g2o, Sophus, Eigen
Perception	PCL, Open3D, OpenCV
Deep Learning	PyTorch, CUDA
Tools	Docker, Git, CMake, Linux